

Edward Zhang

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EDUCATION

Carnegie Mellon University

M.S. in Information Networking

Aug 2025 – May 2027

GPA: 3.71/4.0

University of Southern California

B.S. in Computer Science

Aug 2021 – Dec 2024

GPA: 3.90/4.0

EXPERIENCE

Google

Incoming Software Engineer Intern

May 2026 – Aug 2026

Sunnyvale, CA

Lee's Lab at USC Keck School of Medicine

Software Engineer Intern

Aug 2024 – Dec 2024

Los Angeles, CA

- Developed a web-based DNA sequence analysis platform for HIV research that processes approximately **150-200GB** of genetic data daily across 6 research labs with 30+ users, integrating with researchers' existing Google Drive workflow and eliminating the need for researchers to learn new storage systems.
- Built **React-based** data upload interface using **AWS Lambda** and **SQS** to handle batch processing of multiple 10GB+ files simultaneously with up to **1000 concurrent executions** and implemented critical features including processing status updates and email notifications via **SES**, resulting in **4 new lab adoptions in 2 months**.
- Designed file-scoped **OAuth 2.0** authentication flow where users authorize access to individual Google Drive files on-demand, preventing broad Drive permissions and removing server-side token persistence requirements.

CNeutral.io

Software Engineer Intern

May 2024 – Jul 2024

Singapore

- Developed RESTful APIs using **Python Django** to query Environmental, Social, and Governance (ESG) data from databases containing **10,000+** listed companies and exposed comprehensive analytics like carbon emission and employees diversity to frontend App, enabling investors to access ESG insights for investment decisions.
- Built **PostgreSQL** database tables for companies' ESG data storage, implemented database indexing strategies on ticker symbols and date ranges, and applied **Redis caching** layer for frequently accessed ESG metrics, reducing average dashboard loading time by **0.7 seconds**.
- Built responsive dashboard with **React and Chart.js** featuring interactive time-series charts, heat maps, and multi-criteria filters (industry, region, ESG scores) for cross-company ESG comparison.

PROJECTS

x86 Kernel & Hypervisor | C, x86 Assembly, 15-410

Jan 2026 – Apr 2026

- Built a uniprocessor preemptive multithreaded **x86 kernel** from scratch with **2-level paged virtual memory**; implemented **25+ syscalls** including fork, thread_fork, exec, vanish, wait, and swexn with zero-fill-on-demand and copy-on-write.
- Designed **context switching**, **round-robin scheduling**, and **IDT-based interrupt handling**, along with mutexes and condition variables for preemptive execution.
- Implemented **interrupt-driven device drivers** (PIC timer, keyboard, VGA console), **ELF loader**, **physical frame allocator**, and a **user-space thread library** built on the syscall interface.
- Extended into a **paravirtualizing hypervisor** supporting guest kernels via hypercalls, shadow page tables, virtual interrupt injection, and segmentation-based address translation.

Distributed Bitcoin Mining System | Go, Distributed Systems

Oct 2025

- Architected a **distributed Bitcoin Mining System** in Go with **custom Live Sequence Protocol** over UDP to parallelize mining computation across dynamically scalable miner nodes, **reducing processing time from 28 hours to 17 minutes** for 9-digit hash space by distributing workload across 100+ concurrent miners.
- Engineered a scheduler supporting **concurrent client requests** and **dynamic miner join/leave**, partitioning tasks into independently tracked job blocks and aggregating results to determine global minimum hash.
- Built **fault-tolerant architecture** with epoch-based heartbeat detection and exponential backoff retransmission, **reassigning incomplete jobs** from failed miners to maintain mining progress despite network instability.

TECHNICAL SKILLS

Languages: Python, C/C++, Go, x86 Assembly, Java, JavaScript, TypeScript, Swift, Dart, HTML/CSS, Bash

Frameworks & Libraries: Django, Spring Boot, React, Flask, FastAPI, Express.js, Pandas, NumPy, PyTorch

Databases & Tools: PostgreSQL, Redis, Simics, Docker, K8s, AWS (EC2, S3, DynamoDB, EBS, Lambda), Terraform